MULTIPLICITIES OF COMPONENTS IN A FIELD EXTENSION BASE CHANGE ARE ONE OR POWERS OF THE CHARACTERISTIC

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Allowed standard prerequisites (editorial).

- Geometric reducedness of varieties over perfect fields under base change (Varieties, Lemma varieties-lemma-geometrically-reduced, canonical tag 035X). cited at proof lines 312-314. In characteristic zero k is perfect so the base change is reduced and every multiplicity is one.
- Pushdown formula for lengths along a finite free ring map (Algebra, Lemma algebra-lemma-pushdown-module, canonical tag 02M0). cited at proof lines 337-339. Relates the length at a minimal prime of the smaller tensor product to the length at the unique minimal prime above it, scaled by the residue field degree and the rank.
- Noetherianity of the tensor product with a field extension of finitely generated type (Algebra, Lemma algebra-lemma-Noetherian-field-extension, canonical tag 045I). cited at proof lines 318-319. Makes K tensor_k L Noetherian, so its localizations at minimal primes are artinian local rings and lengths are defined.
- Separable-algebraic splitting of field extensions (Algebra, Lemma algebra-lemma-make-separable, canonical tag 030R). cited at proof lines 345-347. Produces the subfield L/k'/k with L/k' separable and k'/k finite purely inseparable.
- Separable extensions are localizations of smooth algebras (Algebra, Lemma algebra-lemma-localization-smooth-separable, canonical tag 037X). cited at proof lines 350-352. Makes the separable part a localization of a smooth k'-algebra.
- Pullback formula for lengths along flat local homomorphisms with matched maximal ideals (Algebra, Lemma algebra-lemma-pullback-module, canonical tag 02M1). cited at proof lines 357-359. Transports the p-power length from the purely inseparable tensor product to the localized ring S.

Source reference. Original lemma 035X: sources/stacks-chow/tags.txt, source lines 4144–4144. Cited at proof lines 312-314: multiplicity one in characteristic zero.

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Source reference. Original lemma 02M0: sources/stacks-chow/tags.txt, source lines 3427–3427. Cited at proof lines 337-339: descends the length statement along purely inseparable extensions.

Source reference. Original lemma 045I: sources/stacks-chow/tags.txt, source lines 5437–5437. Cited at proof lines 318-319: makes K tensor_k L Noetherian so the localizations at minimal primes are artinian.

Source reference. Original lemma 030R: sources/stacks-chow/tags.txt, source lines 3958–3958. Cited at proof lines 345-346: produces the subfield L/k'/k.

Source reference. Original lemma 037X: sources/stacks-chow/tags.txt, source lines 4216–4216. Cited at proof lines 350-352: makes L/k' a smooth localization.

Source reference. Original lemma 02M1: sources/stacks-chow/tags.txt, source lines 3428–3428. Cited at proof lines 357-359: transports the length along R to S, finishing the proof.

Lemma 0.2. *Let K/k be a field extension. Let Z be an integral locally algebraic scheme over k. The multiplicity m_{Z',Z_K} of an irreducible component $Z' \subset Z_K$ is* 1 *or a power of the characteristic of k.*

Proof. If the characteristic of k is zero, then k is perfect and the multiplicity is always 1 since X_K is reduced by Varieties, Lemma 035X. Assume the characteristic of k is $p > 0$. Let L be the function field of Z. Since Z is locally algebraic over k, the field extension L/k is finitely generated. The ring $K \otimes_k L$ is Noetherian (Algebra, Lemma 045I). Translated into algebra, we have to show that the length of the artinian local ring $(K \otimes_k L)_{\mathfrak{q}}$ is a power of p for every minimal prime ideal $\mathfrak{q}$.

Let L'/L be a finite purely inseparable extension, say of degree p^n. Then $K \otimes_k L \subset K \otimes_k L'$ is a finite free ring map of degree p^n which induces a homeomorphism on spectra and purely inseparable residue field extensions. Hence for every minimal prime $\mathfrak{q}$ as above there is a unique minimal prime $\mathfrak{q}' \subset K \otimes_k L'$ lying over it and

$$p^n \text{length}((K \otimes_k L)_{\mathfrak{q}}) = [\kappa(\mathfrak{q}') : \kappa(\mathfrak{q})] \text{length}((K \otimes_k L')_{\mathfrak{q}'})$$

by Algebra, Lemma 02M0 applied to $M = (K \otimes_k L')_{\mathfrak{q}'} \cong (K \otimes_k L)_{\mathfrak{q}}^{\oplus p^n}$. Since $[\kappa(\mathfrak{q}') : \kappa(\mathfrak{q})]$ is a power of p we conclude that it suffices to prove the statement for L' and $\mathfrak{q}'$.

By the previous paragraph and Algebra, Lemma 030R we may assume that we have a subfield $L/k'/k$ such that L/k' is separable and k'/k is finite purely inseparable. Then $K \otimes_k k'$ is an Artinian local ring. The argument of the preceding paragraph (applied to $L = k$ and $L' = k'$) shows that $\text{length}(K \otimes_k k')$ is a power of p. Since L/k' is the localization of a smooth k'-algebra (Algebra, Lemma 037X). Hence $S = (K \otimes_k L)_{\mathfrak{q}}$ is the localization of a smooth $R = K \otimes_k k'$-algebra at a minimal prime. Thus $R \to S$ is a flat local homomorphism of Artinian local rings and $\mathfrak{m}_R S = \mathfrak{m}_S$. It follows from Algebra, Lemma 02M1 that $\text{length}(K \otimes_k k') = \text{length}(R) = \text{length}(S) = \text{length}((K \otimes_k L)_{\mathfrak{q}})$ and the proof is finished. □